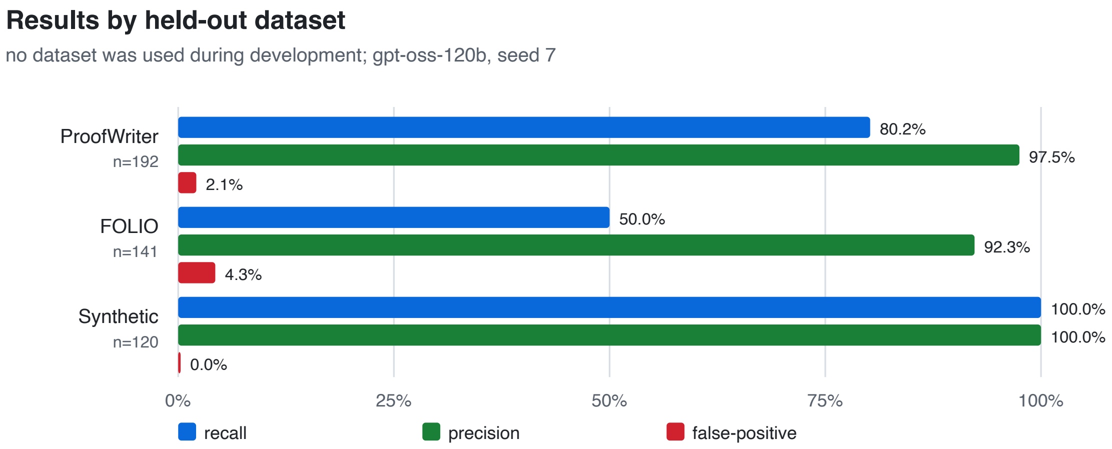
Evaluation

Inconsistency Checker · v0.8.8 · 2026-08-17 · fixed seed 7

The system extracts claims from a document, translates each into first-order logic, and gives every logical judgement to the Z3 solver. It reports the *minimal* set of statements that cannot all be true, together with a derivation of the conflict.

Evaluated on three held-out datasets which none was used during development. Two are externally authored with external labels; the third is generated so its labels follow from construction rather than from inspection.

Headline results



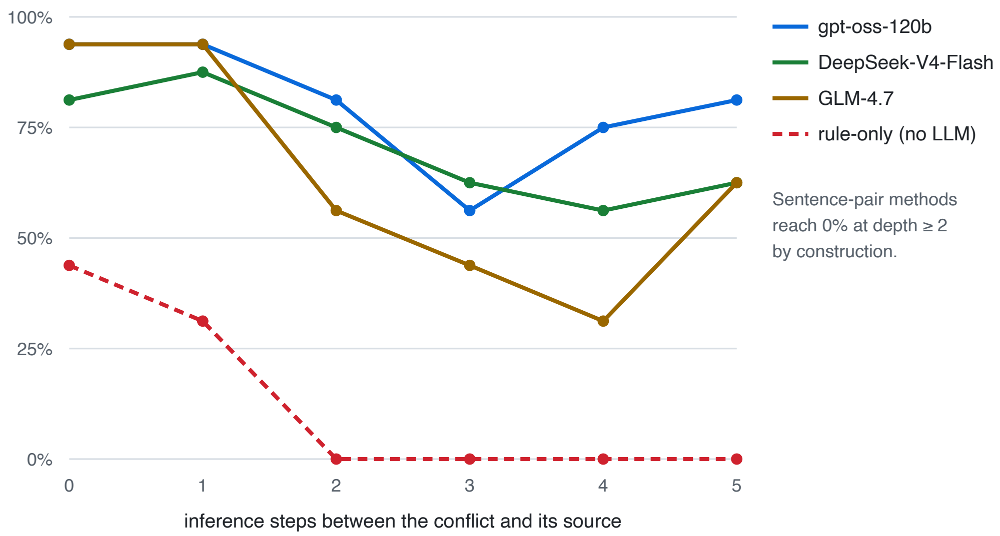
Dataset	n	Source	Recall (95% CI)	Precision	False positives
ProofWriter	192	external	**80.2%** [71.1–87.0]	**97.5%**	**2.1%**
FOLIO	141	external	50.0% [38.8–61.3]	92.3%	4.3%
Synthetic	120	constructed	100% [94.0–100]	100%	0.0%

Balanced by construction (ProofWriter 96/96, FOLIO 72/69, synthetic 60/60), so recall and false-positive rate are measured on equal numbers of positive and negative documents. Intervals are Wilson score intervals.

Recall does not degrade with reasoning depth

Recall vs. reasoning depth

ProofWriter (192 held-out documents, externally authored)



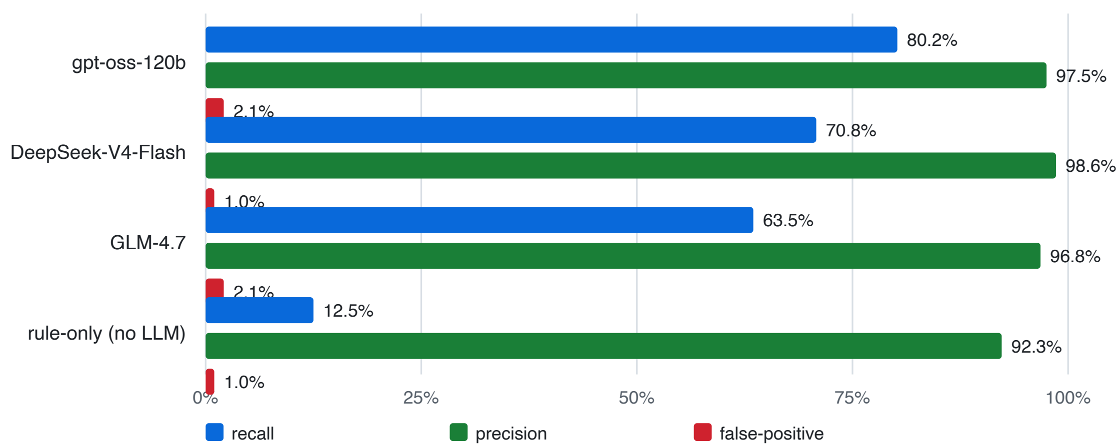
The central claim. A contradiction distributed across *k* inference steps is invisible to any method that compares sentences pairwise: at $k \geq 2$ **no pair of sentences is inconsistent**, so pairwise entailment scores 0% by construction. The full pipeline holds 56–81% through depth 5.

The dashed line is the same pipeline with the language model removed from translation (§ ablation). It collapses to 0% beyond depth 1.

Model comparison and the LLM's contribution

Model comparison

ProofWriter, 192 documents, identical pipeline and seed



Model	Recall	Precision	FP	Tokens/doc	s/doc
gpt-oss-120b	**80.2%**	97.5%	2.1%	5,168	**6.9**
DeepSeek-V4-Flash	70.8%	**98.6%**	**1.0%**	**4,924**	59.1
GLM-4.7	63.5%	96.8%	2.1%	18,381	31.6
rule-only (LLM withheld)	*12.5%*	*92.3%*	*1.0%*	*0*	*offline*

Ablation. Withholding the LLM translator while leaving extraction, the gate, the solver and reporting untouched drops recall from **80.2%** → **12.5%** (and 100% → 1.7% on the synthetic set). Recall is therefore bounded by *translation coverage*, not by the reasoning engine — the deterministic translator currently produces usable logic for about 5% of real sentences. This is the primary target for future work.

Precision never falls below 92% in any configuration, including with no LLM at all. The system does not manufacture contradictions.

Model recall differences have overlapping confidence intervals and should be read as suggestive, not established. The models were not run under identical settings: gpt-oss cannot disable internal reasoning on its provider, DeepSeek can.

Method

Every verdict is a Z3 proof, not a model judgement. The language model only proposes formulas. Content outside the supported fragment (modality, tense, arithmetic, comparatives, hedged generalisations) is **quarantined with a written reason** rather than guessed at; quarantine rates were 3.2% (ProofWriter), 7.5% (FOLIO), 2.7% (synthetic). ProofWriter and FOLIO are entailment datasets, converted to consistency checking by appending the negated conclusion: a theory entails Q exactly when the theory together with $\neg Q$ is inconsistent. The conversion rule was fixed before any result was seen.

Reproducibility. Fixed seed and temperature 0; the full test suite (255 tests) runs offline against shipped fixtures with no API key. Datasets, adapters, scorer and charts are in `tools/`; per-document outputs and `metrics.json` for every run are retained.

Limitations. The synthetic set was authored by us and exercises the fragment the system targets; the external results are the load-bearing ones. The depth-3 dip is non-monotonic across all three models and is a sampling artefact at 16 documents per cell, not a depth effect. Performance tracks how *logical* a contradiction is: conflicts resting on arithmetic, dates, degree or world knowledge are outside the representation by design.